

Filtering numbers

INTERMEDIATE SQL



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WHERE

- **WHERE** filtering clause



WHERE

```
WHERE color = 'green'
```



WHERE with comparison operators

```
SELECT title
FROM films
WHERE release_year > 1960;
```

```
|title|
|-----|
|Judgment at Nuremberg|
|Pocketful of Miracles|
|The Hustler|
|The Misfits|
|...|
```

Comparison operators

```
SELECT title
FROM films
WHERE release_year < 1960;
```

```
|title|
|-----|
|Intolerance:Love's Struggle Throughout the Ages|
|Over the Hill to the Poorhouse|
|The Big Parade|
|Metropolis|
|...|
```

Comparison operators

```
SELECT title
FROM films
WHERE release_year <= 1960;
```

```
|title|
|-----|
|Intolerance:Love's Struggle Throughout the Ages|
|Over the Hill to the Poorhouse|
|The Big Parade|
|Metropolis|
|...|
```

Comparison operators

```
SELECT title
FROM films
WHERE release_year = 1960;
```

```
|title      |
|-----|
|Elmer Gantry|
|Psycho     |
|The Apartment|
```


Comparison operators

```
SELECT title
FROM films
WHERE release_year <> 1960;
```

```
|title|
|-----|
|Intolerance:Love's Struggle Throughout the Ages|
|Over the Hill to the Poorhouse|
|The Big Parade|
|Metropolis|
|...|
```


Comparison operators

- `>` Greater than or after
- `<` Less than or before
- `=` Equal to
- `>=` Greater than or equal to
- `<=` Less than or equal to
- `<>` Not equal to

WHERE with strings

- Use single-quotes around strings we want to filter

```
SELECT title
FROM films
WHERE country = 'Japan';
```

```
|title          |
|-----|
|Seven Samurai |
|Tora! Tora! Tora!|
|Akira          |
|Madadayo       |
|Street Fighter |
|...            |
```

Order of execution

```
-- Written code:  
SELECT item  
FROM coats  
WHERE color = 'green'  
LIMIT 5;
```

- Order of execution:

- **FROM** coats
- **WHERE** color = 'green'
- **SELECT** item
- **LIMIT** 5;

Let's practice!
INTERMEDIATE SQL

Multiple criteria

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Multiple criteria



Multiple criteria



Multiple criteria



Multiple criteria

- OR , AND , BETWEEN

```
SELECT *  
FROM coats  
WHERE color = 'yellow' OR length = 'short';
```

```
SELECT *  
FROM coats  
WHERE color = 'yellow' AND length = 'short';
```

```
SELECT *  
FROM coats  
WHERE buttons BETWEEN 1 AND 5;
```

OR operator

- Use `OR` when you need to satisfy at least one condition



OR operator

- Correct:

```
SELECT title
FROM films
WHERE release_year = 1994
      OR release_year = 2000;
```

```
|title|
|-----|
|3 Ninjas Kick Back|
|A Low Down Dirty Shame|
|Ace Ventura:Pet Detective|
|...|
```

- Invalid:

```
SELECT title
FROM films
WHERE release_year = 1994 OR 2000;
```

```
argument of OR must be type boolean,
not type integer
LINE 3: WHERE release_year = 1994
OR 2000;
      ^
```

AND operator

- Use `AND` if we need to satisfy all criteria
- Correct:

```
SELECT title
FROM films
WHERE release_year > 1994
      AND release_year < 2000;
```

```
|title|
|-----|
|Ace Ventura:When Nature Calls|
|Apollo 13|
|Assassins|
|Babe|
|...|
```

- Invalid:

```
SELECT title
FROM films
WHERE release_year > 1994 AND < 2000;
```

```
syntax error at or near "[removed]"
1994 AND < 2000;
      ^
```

AND, OR

- Filter films released in 1994 or 1995, and certified PG or R
- Enclose individual clauses in parentheses

```
SELECT title
FROM films
WHERE (release_year = 1994 OR release_year = 1995)
      AND (certification = 'PG' OR certification = 'R');
```

```
|title|
|-----|
|3 Ninjas Kick Back|
|A Low Down Dirty Shame|
|Baby's Day Out|
|Beverly Hills Cop III|
|...|
```

BETWEEN, AND

```
SELECT title
FROM films
WHERE release_year >= 1994
      AND release_year <= 2000;
```

```
|title|
|-----|
|3 Ninjas Kick Back|
|A Low Down Dirty Shame|
|Ace Ventura:Pet Detective|
|Baby's Day Out|
|...|
```

```
SELECT title
FROM films
WHERE release_year
      BETWEEN 1994 AND 2000;
```

```
|title|
|-----|
|3 Ninjas Kick Back|
|A Low Down Dirty Shame|
|Ace Ventura:Pet Detective|
|Baby's Day Out|
|...|
```

BETWEEN, AND, OR

```
SELECT title
FROM films
WHERE release_year
BETWEEN 1994 AND 2000 AND country='UK';
```

```
|title|
|-----|
|Four Weddings and a Funeral|
|The Hudsucker Proxy|
|Dead Man Walking|
|GoldenEye|
|...|
```


Let's practice!
INTERMEDIATE SQL

Filtering text

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Filtering text

- `WHERE` can also filter text

```
SELECT title
FROM films
WHERE country = 'Japan';
```

```
|title|
|-----|
|Seven Samurai|
|Tora! Tora! Tora!|
|Akira|
|Madadayo|
|Street Fighter|
```

Filtering text

- Filter a pattern rather than specific text
- `LIKE`
- `NOT LIKE`
- `IN`

LIKE

- Used to search for a pattern in a field

% match zero, one, or many characters

```
SELECT name
FROM people
WHERE name LIKE 'Ade%';
```

name	

AdeL Karam	
Adelaide Kane	
Aden Young	

_ match a single character

```
SELECT name
FROM people
WHERE name LIKE 'Ev_';
```

name	

Eve	

- Ev_ Mendes

NOT LIKE

```
SELECT name
FROM people;
```

```
| name |
|-----|
| 50 Cent |
| A. Michael Baldwin |
| A. Raven Cruz |
| A.J. Buckley |
| A.J. DeLucia |
| ... |
```

```
SELECT name
FROM people
WHERE name NOT LIKE 'A.%';
```

```
| name |
|-----|
| 50 Cent |
| Aaliyah |
| Aaron Ashmore |
| Aaron Hann |
| ... |
```

Wildcard position

```
SELECT name
FROM people
WHERE name LIKE '%r';
```

```
|name|
|-----|
|A.J. Langer|
|Aaron Schneider|
|Aaron Seltzer|
|Abigail Spencer|
|...|
```

```
SELECT name
FROM people
WHERE name LIKE '__t%';
```

```
|name|
|-----|
|Aitana Sánchez-Gijón|
|Anthony 'Critic' Campos|
|Anthony Bell|
|Anthony Burrell|
|...|
```

WHERE, OR

```
SELECT title
FROM films
WHERE release_year = 1920
OR release_year = 1930
OR release_year = 1940;
```

```
|title|
|-----|
|Over the Hill to the Poorhouse|
|Hell's Angels|
|Boom Town|
|...|
```


WHERE, IN

```
SELECT title
FROM films
WHERE release_year IN (1920, 1930, 1940);
```

```
|title|
|-----|
|Over the Hill to the Poorhouse|
|Hell's Angels|
|Boom Town|
|...|
```


WHERE, IN

```
SELECT title
FROM films
WHERE country IN ('Germany', 'France');
```

```
|title      |
|-----|
|Metropolis|
|Pandora's Box|
|The Train |
|...       |
```

Let's practice!
INTERMEDIATE SQL

NULL values

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Missing values

- `COUNT(field_name)` includes only non-missing values
- `COUNT(*)` includes missing values

`null`

- Missing values:
 - Human error
 - Information not available
 - Unknown

null

```
SELECT COUNT(*) AS count_records  
FROM people;
```

```
|count_records|  
|-----|  
|8397        |
```

```
SELECT *  
FROM people;
```

```
|id|name                |birthdate |deathdate|  
|--|-----|-----|-----|  
|1 |50 Cent              |1975-07-06|null    |  
|2 |A. Michael Baldwin  |1963-04-04|null    |  
|3 |A. Raven Cruz       |null      |null    |  
... 
```

IS NULL

```
SELECT name  
FROM people  
WHERE birthdate IS NULL;
```

```
|name          |  
|-----|  
|A. Raven Cruz|  
|A.J. DeLucia |  
|Aaron Hann   |  
|...          |
```

IS NOT NULL

```
SELECT COUNT(*) AS no_birthdates
FROM people
WHERE birthdate IS NULL;
```

```
|no_birthdates|
|-----|
|2245        |
```

```
SELECT COUNT(name) AS count_birthdates
FROM people
WHERE birthdate IS NOT NULL;
```

```
|count_birthdates|
|-----|
|6152            |
```


COUNT() vs IS NOT NULL

```
SELECT
  COUNT(certification)
  AS count_certification
FROM films;
```

```
|count_certification|
|-----|
|4666              |
```

```
SELECT
  COUNT(certification)
  AS count_certification
FROM films
WHERE certification IS NOT NULL;
```

```
|count_certification|
|-----|
|4666              |
```


NULL put simply

- `NULL` values are missing values
- Very common
- Use `IS NULL` or `IS NOT NULL` to:
 - Identify missing values
 - Select missing values
 - Exclude missing values

Let's practice!
INTERMEDIATE SQL